Input layer: the genes

every gene $i = 1, \ldots, N$ (N = 6,607):

x_i = [pathway membership (173) ‖ ProtT5 (1024)] $\in R^{1197}$

preprocessor and token stack in: $x_1, \ldots, x_N$

$h_i^{(0)} = \mathrm{LN}(W_0 x_i + b_0) \in R^{12}$

out: $H^{(0)} = [\, c;\ h_1^{(0)};\ \ldots;\ h_N^{(0)} \,] \in R^{(N+1)\times 12}$,

c a learned class token; 14,400 + 12 parameters

transformer block (4 heads, head width 3) in: $H^{(0)}$

$\alpha^{(k)} = \mathrm{softmax}(Q_k K_k^{\top}/\sqrt{3})$: every token attends to all $N + 1$

$Z = \mathrm{LN}(H^{(0)} + W_O [\, \alpha^{(1)} V_1;\ \ldots;\ \alpha^{(4)} V_4 \,])$

$H^{(1)} = \mathrm{LN}(Z + \mathrm{FFN}(Z))$

out: $H^{(1)} \in R^{(N+1)\times 12}$; 1,884 parameters

no graph prior here (the KL pull or the mask act on a second block)

input: the strain, one deleted gene g, $S = \{g\}$

deletion operator (cross-attention, 6 heads)

in: $H = H^{(1)}_{1:N}$ (one row per gene), S

gene i queries the deleted gene: $c_i = \mathrm{Attn}(h_i;\ \{h_p : p \in S\})$

$|S| = 1 \Rightarrow c_i = W_O(W_V h_g + b_V) + b_O$, the same for every i

$Z_i = \mathrm{LN}(h_i + c_i)$, $h_i^{\mathrm{pert}} = \mathrm{LN}(Z_i + \mathrm{FFN}(Z_i))$

out: $H^{\mathrm{pert}} \in R^{N\times 12}$; 1,884 parameters

input: the metabolic solve of the deletion of g

$u_g = [\, r_g;\ \log(1 + |v_g|) \,] \in R^{4135}$

$r_g \in [0,1]^4$ the FBA growth ratios on four media,

$v_g \in R^{4131}$ the pFBA fluxes on YPD

readout in: $h_{\mathrm{CLS}} = H_0^{(1)}$ (the reference cell), H^{pert}, S, u_g

$z = [\, h_{\mathrm{CLS}};\ \frac{1}{N}\sum_i h_i^{\mathrm{pert}};\ \frac{1}{|S|}\sum_{p\in S} h_p^{\mathrm{pert}} \,] \in R^{36}$

$\ell_g = w_2^{\top}\,\mathrm{ReLU}(W_1 z + b_1) + b_2 \ + \ w^{\top} u_g + b$

out: $\ell_g \in R$; 457 + 4,136 parameters

in: ℓ_g; out: essential if $\ell_g > 0$

class-weighted BCE on 629 training genes, 200 epochs of AdamW;

read at the best validation epoch, scored on 195 held-out genes